Lab: Do the bees at hedgerows and weedy field margins (controls) have the same distribution of traits?

Our goal is to reproduce the analyses in Ponisio et al. 2016, On-farm habitat restoration counters biotic homogenization in intensively managed agriculture, Ponisio et al. 2016

Specifically, we are going to reproduce Figure 4 panels a-b, where we examine whether the bees at hedgerows (restored habitat) and controls (weedy field margins that would have a hedgerow) are random subset of the species pool.

Conservation/ecology Topics

- Become familiar with different metrics of biodiversity, including trait diversity.
- Become familiar with the concept of biotic homogenization.

Computational Topics

- Join data
- Make histograms
- Subsetting, reshaping data using dplyr (more practice)
- Use logical indexing to subset data (more practice)
- Write functions (more practice)
- Write for loops (more practice)

Statistical Topics

- AB testing, simulating the null hypothesis
- Calculate empirical p-values
- Hypothesis testings: reject or accept our null hypothesis

```r
# Load necessary libraries in R
library(tidyverse)

# For reproducibility purposes
set.seed(123)
```

```r
## read in the specimen data
spec <- read.csv("data/specimens-complete.csv")
```

```
## Warning in file(file, "rt"): cannot
## open file
## 'data/specimens-complete.csv': No such
## file or directory
```

```
## Error in file(file, "rt"): cannot open the connection
```

```r
## explore the columns
str(spec)
```

```
## 'data.frame':    8810 obs. of  5 variables:
##  $ UniqueID    : chr  "M2007SR1Barger_002" "M2007SR1Barger_004" "M2007SR1Barger_008" "M2007SR1Barger_
##  $ GenusSpecies: chr  "Lasioglossum (Dialictus) incompletum" "Ceratina acantha" "Halictus ligatus" "
##  $ Site        : chr  "Barger" "Barger" "Barger" "Barger" ...
##  $ SiteStatus  : chr  "control" "control" "control" "control" ...
##  $ Year        : int  2007 2007 2007 2007 2007 2007 2007 2007 2007 2007 ...
```

Each row is a uniquely identified insect specimen and all of it's attributes. If we explore the species ID-related columns in more depth, we can see that there are many different types of insects in addition to bees. There are also some specimens that have never been IDed (missed data is a blank cell in these case).

```r
unique(spec$Order)
```

```
## NULL
```

```r
unique(spec$Family)
```

```
## NULL
```

```r
sort(unique(spec$GenusSpecies))
```

```
##  [1] "Agapostemon texanus"
##  [2] "Andrena angustitarsata"
##  [3] "Andrena auricoma"
##  [4] "Andrena candida"
##  [5] "Andrena chlorogaster"
##  [6] "Andrena cressonii infasciata"
##  [7] "Andrena knuthiana"
##  [8] "Andrena nigrocaerulea"
##  [9] "Andrena orthocarpi"
## [10] "Andrena piperi"
## [11] "Andrena scurra"
## [12] "Andrena subaustralis"
## [13] "Andrena subchalybea"
## [14] "Anthidiellum notatum robertsoni"
## [15] "Anthophora urbana"
## [16] "Ashmeadiella aridula astragali"
## [17] "Ashmeadiella bucconis denticulata"
## [18] "Ashmeadiella cactorum basalis"
## [19] "Bombus californicus"
## [20] "Bombus melanopygus"
## [21] "Bombus vosnesenskii"
## [22] "Calliopsis fracta"
## [23] "Calliopsis hesperia equina"
## [24] "Calliopsis scutellaris"
## [25] "Ceratina acantha"
## [26] "Ceratina arizonensis"
## [27] "Ceratina dallatorreana"
## [28] "Ceratina nanula"
## [29] "Ceratina punctigena"
## [30] "Coelioxys novomexicanus"
## [31] "Colletes hyalinus"
```

```
## [32] "Diadasia consociata"
## [33] "Diadasia enavata"
## [34] "Eucera actuosa"
## [35] "Halictus ligatus"
## [36] "Halictus tripartitus"
## [37] "Hoplitis producta gracilis"
## [38] "Hylaeus calvus"
## [39] "Hylaeus conspicuus"
## [40] "Hylaeus leptocephalus"
## [41] "Hylaeus mesillae"
## [42] "Hylaeus rudbeckiae"
## [43] "Lasioglossum (Dialictus) brunneiventre"
## [44] "Lasioglossum (Dialictus) diversopunctatum"
## [45] "Lasioglossum (Dialictus) impavidum"
## [46] "Lasioglossum (Dialictus) incompletum"
## [47] "Lasioglossum (Dialictus) knereri"
## [48] "Lasioglossum (Dialictus) punctatoventre"
## [49] "Lasioglossum (Dialictus) tegulare group"
## [50] "Lasioglossum (Evylaeus) diatretum"
## [51] "Lasioglossum (Evylaeus) granosum"
## [52] "Lasioglossum (Evylaeus) kincaidii"
## [53] "Lasioglossum (Evylaeus) nigrescens"
## [54] "Lasioglossum (Evylaeus) sp. B"
## [55] "Lasioglossum mellipes"
## [56] "Lasioglossum sisymbrii"
## [57] "Lasioglossum titusi"
## [58] "Megachile angelarum"
## [59] "Megachile apicalis"
## [60] "Megachile brevis"
## [61] "Megachile coquilletti"
## [62] "Megachile fidelis"
## [63] "Megachile frugalis"
## [64] "Megachile gentilis"
## [65] "Megachile lippiae"
## [66] "Megachile montivaga"
## [67] "Megachile onobrychidis"
## [68] "Megachile prosopidis"
## [69] "Megachile rotundata"
## [70] "Melissodes agilis"
## [71] "Melissodes lupinus"
## [72] "Melissodes robustior"
## [73] "Melissodes stearnsi"
## [74] "Melissodes tepida timberlakei"
## [75] "Nomada sp. 3"
## [76] "Nomada sp. A"
## [77] "Nomada sp. CK-1"
## [78] "Osmia aglaia"
## [79] "Osmia atrocyanea"
## [80] "Osmia gaudiosa"
## [81] "Osmia laeta"
## [82] "Osmia lignaria propinqua"
## [83] "Osmia nemoris"
## [84] "Osmia regulina"
## [85] "Peponapis pruinosa"
```

```
## [86] "Sphecodes sp. B"
## [87] "Svastra obliqua expurgata"
## [88] "Triepeolus concavus"
## [89] "Triepeolus heterurus"
## [90] "Triepeolus melanarius"
## [91] "Triepeolus sp. A"
## [92] "Triepeolus subnitens"
## [93] "Triepeolus timberlakei"
## [94] "Triepeolus utahensis"
## [95] "Xeromelecta californica"
## [96] "Xylocopa sonorina"
## [97] "Xylocopa tabaniformis orpifex"
```

Question 1: Practicing the data subsetting tools from dplyr

- 1a. Thinking back to the dplyr tools you learned last week, subset the data to only bees. The families of bees are "Halictidae", "Megachilidae", "Apidae", "Colletidae", "Andrenidae" and "Melittidae". In addition, drop all rows without anything in the 'Species' column, otherwise we could end up with specimens only identified to Genus.

Keep the data object named spec (i.e., don't create a new name for the subsetted data.)

HINT: %in% maybe useful, or remember how to string together Booleans in 'dplyr'.

```r
library(dplyr)

spec <- spec %>%
  filter(
    Family %in% c("Halictidae", "Megachilidae", "Apidae",  "Colletidae", "Andrenidae", "Melittidae"),
    !is.na(Species),
    Species != ""
)
```

```
## Error in `filter()`:
## i In argument: ``%in%`(...)`.
## Caused by error:
## ! object 'Family' not found
```

```r
### The dimensions of the subsetted data should be 99097 by 40
# dim(spec)
```

- 1b. Now take a look at the site statuses, or the site "treatments",

```r
unique(spec$SiteStatus)
```

```
## [1] "control" "mature"
```

This data include many more site treatments than we need to reproduce the publication because it includes all of the sites ever surveyed in this landscape. We don't really care about the restored sites (that is the year the hedgerow was planted so they are between a hedgerow and a control). We don't have many "natural" sites (we couldn't find any). Forbs are hedgerows with additional forb plantings, also not what we need.

- 1c. Subset only to mature hedgerows and controls (weedy field margins). We also only care about the "hand-netted" data (the column 'NetPan') and female bees ('Sex') (male bees don't collect pollen). Do all of this subsetting in one call to filter (you will need to combine your comparisons using & or |.

Similar to before keep the name of the data the same. The publication also includes the maturing sites, but we are going to make our lives a little easier by keeping our analysis to two treatments.

```
spec <- spec %>%
  filter(
    SiteStatus %in% c("control", "mature"),
    NetPan == "net",
    Sex == "f"
  )
```

```
## Error in `filter()`:
## i In argument: `NetPan ==
##   "net"`.
## Caused by error:
## ! object 'NetPan' not found
```

```
### The dimensions of the subsetted data should be 8810 by 40.
# dim(spec)
```

- 1d. Lastly, drop all the random columns we don't care about. All we want to keep are "UniqueID", "GenusSpecies", "Site", "SiteStatus" and "Year".

```
### let's drop all the random columns we don't care about
spec <- spec %>%
  select("UniqueID","GenusSpecies","Site", "SiteStatus", "Year")
```

Demo: Joining data

Next, we want to add in species traits.

```
### load the traits table, this includes traits for all the bees
traits <- read.csv('data/bee.csv')
```

```
## Warning in file(file, "rt"): cannot
## open file 'data/bee.csv': No such file
## or directory
```

```
## Error in file(file, "rt"): cannot open the connection
```

We will often find ourselves in a situation where we want to combine two datasets by some shared column. In dyplyr, mutating joins add columns from y to x, matching observations based on the keys. There are four mutating joins: the inner join, and the three outer joins.

- Inner join: An inner_join() only keeps observations from x that have a matching key in y. The most important property of an inner join is that unmatched rows in either input are not included in the result, so data is dropped.

```r
df1 <- data.frame(x = c(1:4, NA), y = 2)
df2 <- data.frame(x = c(1:2), z = 3)
head(df1)
```

```
##    x y
## 1  1 2
## 2  2 2
## 3  3 2
## 4  4 2
## 5 NA 2
```

```r
head(df2)
```

```
##   x z
## 1 1 3
## 2 2 3
```

```r
inner <- inner_join(df1, df2)
```

```
## Joining with `by = join_by(x)`
```

```r
dim(inner)
```

```
## [1] 2 3
```

```r
head(inner)
```

```
##   x y z
## 1 1 2 3
## 2 2 2 3
```

- Outer joins: The three outer joins keep observations that appear in at least one of the data frames:
- A left_join() keeps all observations in x.

```r
df1 <- data.frame(x = c(1:100), y = 2)
df2 <- data.frame(x = c(1:10), z = 3)
head(df1)
```

```
##   x y
## 1 1 2
## 2 2 2
## 3 3 2
## 4 4 2
## 5 5 2
## 6 6 2
```

```r
head(df2)
```

```
##   x z
## 1 1 3
## 2 2 3
## 3 3 3
## 4 4 3
## 5 5 3
## 6 6 3
```

```r
left <- left_join(df1, df2)
```

```
## Joining with `by = join_by(x)`
```

```r
dim(left)
```

```
## [1] 100   3
```

```r
head(left)
```

```
##   x y z
## 1 1 2 3
## 2 2 2 3
## 3 3 2 3
## 4 4 2 3
## 5 5 2 3
## 6 6 2 3
```

- A right_join() keeps all observations in y.

```r
df1 <- data.frame(x = c(1:100), y = 2)
df2 <- data.frame(x = c(1:10), z = 3)
head(df1)
```

```
##   x y
## 1 1 2
## 2 2 2
## 3 3 2
## 4 4 2
## 5 5 2
## 6 6 2
```

```r
head(df2)
```

```
##   x z
## 1 1 3
## 2 2 3
## 3 3 3
## 4 4 3
## 5 5 3
## 6 6 3
```

```r
right <- right_join(df1, df2)
```

```
## Joining with `by = join_by(x)`
```

```r
dim(right)
```

```
## [1] 10  3
```

```r
head(right)
```

```
##   x y z
## 1 1 2 3
## 2 2 2 3
## 3 3 2 3
## 4 4 2 3
## 5 5 2 3
## 6 6 2 3
```

- A full_join() keeps all observations in x and y.

```r
df1 <- data.frame(x = c(1:100), y = 2)
df2 <- data.frame(x = c(1:10, 100:200), z = 3)
df1
```

```
##       x y
## 1     1 2
## 2     2 2
## 3     3 2
## 4     4 2
## 5     5 2
## 6     6 2
## 7     7 2
## 8     8 2
## 9     9 2
## 10   10 2
## 11   11 2
## 12   12 2
## 13   13 2
## 14   14 2
## 15   15 2
## 16   16 2
## 17   17 2
## 18   18 2
## 19   19 2
## 20   20 2
## 21   21 2
## 22   22 2
## 23   23 2
## 24   24 2
## 25   25 2
```

```
## 26    26 2
## 27    27 2
## 28    28 2
## 29    29 2
## 30    30 2
## 31    31 2
## 32    32 2
## 33    33 2
## 34    34 2
## 35    35 2
## 36    36 2
## 37    37 2
## 38    38 2
## 39    39 2
## 40    40 2
## 41    41 2
## 42    42 2
## 43    43 2
## 44    44 2
## 45    45 2
## 46    46 2
## 47    47 2
## 48    48 2
## 49    49 2
## 50    50 2
## 51    51 2
## 52    52 2
## 53    53 2
## 54    54 2
## 55    55 2
## 56    56 2
## 57    57 2
## 58    58 2
## 59    59 2
## 60    60 2
## 61    61 2
## 62    62 2
## 63    63 2
## 64    64 2
## 65    65 2
## 66    66 2
## 67    67 2
## 68    68 2
## 69    69 2
## 70    70 2
## 71    71 2
## 72    72 2
## 73    73 2
## 74    74 2
## 75    75 2
## 76    76 2
## 77    77 2
## 78    78 2
## 79    79 2
```

```
## 80   80 2
## 81   81 2
## 82   82 2
## 83   83 2
## 84   84 2
## 85   85 2
## 86   86 2
## 87   87 2
## 88   88 2
## 89   89 2
## 90   90 2
## 91   91 2
## 92   92 2
## 93   93 2
## 94   94 2
## 95   95 2
## 96   96 2
## 97   97 2
## 98   98 2
## 99   99 2
## 100 100 2
```

```
df2
```

```
##       x z
## 1     1 3
## 2     2 3
## 3     3 3
## 4     4 3
## 5     5 3
## 6     6 3
## 7     7 3
## 8     8 3
## 9     9 3
## 10   10 3
## 11  100 3
## 12  101 3
## 13  102 3
## 14  103 3
## 15  104 3
## 16  105 3
## 17  106 3
## 18  107 3
## 19  108 3
## 20  109 3
## 21  110 3
## 22  111 3
## 23  112 3
## 24  113 3
## 25  114 3
## 26  115 3
## 27  116 3
## 28  117 3
## 29  118 3
```

```
## 30  119 3
## 31  120 3
## 32  121 3
## 33  122 3
## 34  123 3
## 35  124 3
## 36  125 3
## 37  126 3
## 38  127 3
## 39  128 3
## 40  129 3
## 41  130 3
## 42  131 3
## 43  132 3
## 44  133 3
## 45  134 3
## 46  135 3
## 47  136 3
## 48  137 3
## 49  138 3
## 50  139 3
## 51  140 3
## 52  141 3
## 53  142 3
## 54  143 3
## 55  144 3
## 56  145 3
## 57  146 3
## 58  147 3
## 59  148 3
## 60  149 3
## 61  150 3
## 62  151 3
## 63  152 3
## 64  153 3
## 65  154 3
## 66  155 3
## 67  156 3
## 68  157 3
## 69  158 3
## 70  159 3
## 71  160 3
## 72  161 3
## 73  162 3
## 74  163 3
## 75  164 3
## 76  165 3
## 77  166 3
## 78  167 3
## 79  168 3
## 80  169 3
## 81  170 3
## 82  171 3
## 83  172 3
```

```
## 84  173 3
## 85  174 3
## 86  175 3
## 87  176 3
## 88  177 3
## 89  178 3
## 90  179 3
## 91  180 3
## 92  181 3
## 93  182 3
## 94  183 3
## 95  184 3
## 96  185 3
## 97  186 3
## 98  187 3
## 99  188 3
## 100 189 3
## 101 190 3
## 102 191 3
## 103 192 3
## 104 193 3
## 105 194 3
## 106 195 3
## 107 196 3
## 108 197 3
## 109 198 3
## 110 199 3
## 111 200 3
```

```r
full <- full_join(df1, df2)
```

```
## Joining with `by = join_by(x)`
```

```r
dim(full)
```

```
## [1] 200   3
```

```r
head(full)
```

```
##   x y z
## 1 1 2 3
## 2 2 2 3
## 3 3 2 3
## 4 4 2 3
## 5 5 2 3
## 6 6 2 3
```

Class discussion: What type of join do we want to use to to combine the individual bee data with their traits? - Left join

Question 2

- 2a. Join the trait data to the individual bee data. Before joining the data, subset the trait to only the bee IDs ('GenusSpecies'), body size ('MeanITD'), floral specialization ('d'). Call the joined data spec.traits

```
### your code here
spec.traits <- spec %>%
  left_join(
    traits %>%
      select(GenusSpecies, MeanITD, d),
    by = "GenusSpecies"
  )

## Observe individual bee data traits
dim(spec.traits)
```

```
## [1] 8810    7
```

- 2b. How many specimens are missing body size data? HINT: the function 'is.na()' may be useful. EXTRA HINT: Remember the TRUEs are counted as 1s in R.

```
### your code here
sum(is.na(spec.traits$MeanITD))
```

```
## [1] 176
```

- 2c. What are the unique species names missing body size data.

```
### your code here
unique(spec.traits$GenusSpecies[is.na(spec.traits$MeanITD)])
```

```
## [1] "Melissodes lupinus"
## [2] "Lasioglossum (Dialictus) knereri"
## [3] "Xylocopa sonorina"
## [4] "Lasioglossum (Dialictus) brunneiventre"
## [5] "Nomada sp. CK-1"
## [6] "Megachile prosopidis"
## [7] "Triepeolus utahensis"
## [8] "Coelioxys novomexicanus"
```

- 2d. These were relatively rare species that didn't have enough individuals to accurately estimate body size. Since we cannot do much with them, go ahead and drop them. Keep the data named spec.traits. You can use the tools we have learned in dyplyr, or just clever indexing.

HINT: There are several ways you can drop the rows without body size data, try to avoid hard coding by copy pasting the names above. What if you added new data and wanted to re-run the code? You might miss some species if you remove them "by hand". EXTRA HINT: Remember that ! reverses TRUEs and FALSEs (so TRUE because FALSE and vice versa).

EXTRA EXTRA HINT: If you decide to use indexing, remember that when we index a 2D object we need to specify the indexes of the rows and columns.

```r
### your code here
spec.traits <- spec.traits[!is.na(spec.traits$MeanITD), ]
### The dimensions should be 8634 7
dim(spec.traits)
```

```
## [1] 8634    7
```

Lab Question 3

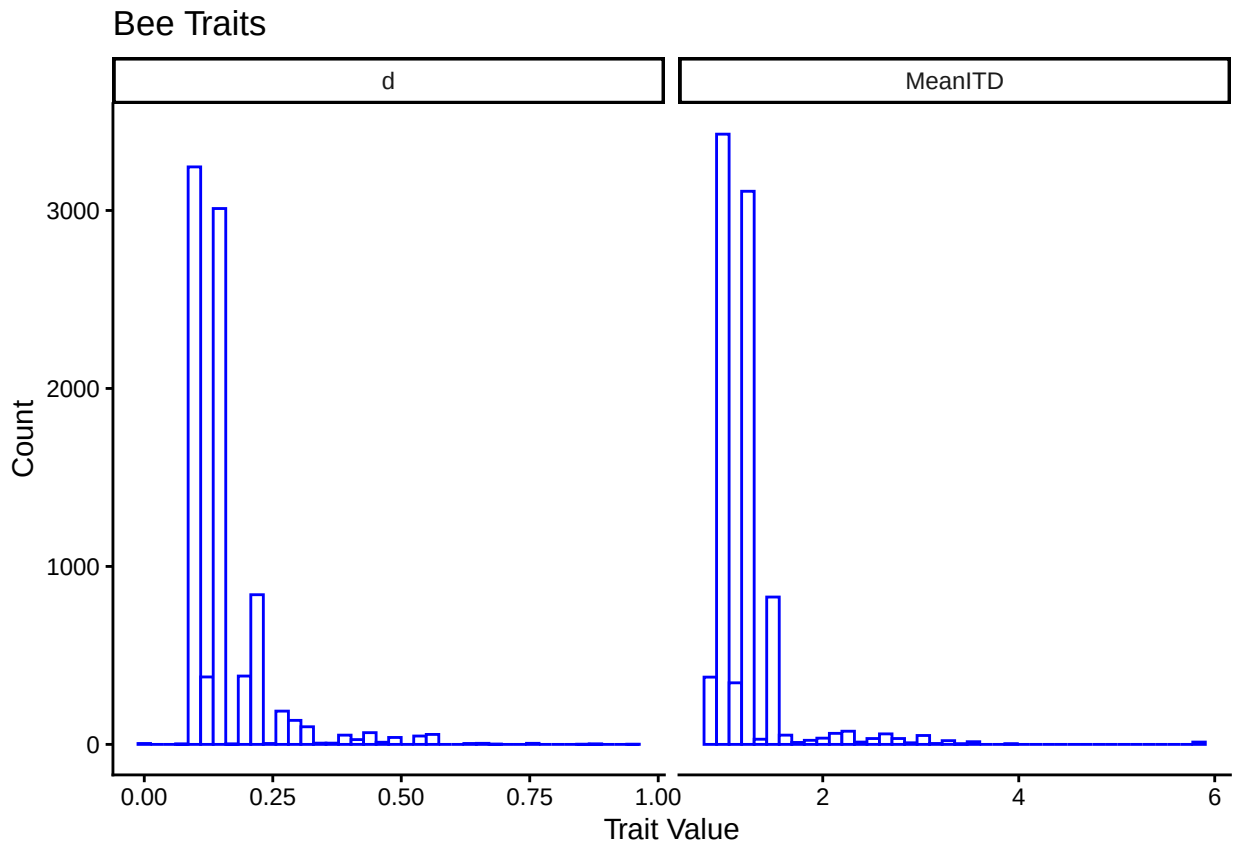
- 3a. Use 'group_by' and 'summarize' to calculate the average body size and specialization for each site, year, site status combination. Create a new object called sys.traits

```r
### your code here
sys.traits <- spec.traits %>%
   group_by(UniqueID,Year, SiteStatus) %>%
summarise(n_mean_bodysize = mean(MeanITD, na.rm = TRUE))
```

```
## `summarise()` has grouped output by
## 'UniqueID', 'Year'. You can override
## using the `.groups` argument.
```

- 3b. Visualize this data with a histograms for each trait. HINT: 'geom_histogram()' is the function that makes histograms, and it only needs an x aesthetic.

```r
### body size
### your code here
library(tidyverse)
spec.traits %>%
pivot_longer(
  cols = c(MeanITD, d),
names_to = "Trait",
values_to = "Value"
) %>%
  ggplot(aes(x = Value)) +
geom_histogram(color = "blue", fill = "white", bins = 40) +
  facet_wrap(~ Trait, scales = "free_x") +
  theme_classic() +
  labs(title = "Bee Traits",
x = "Trait Value",
y = "Count"
)
```




```
### Specialization
spec.traits %>%
  group_by(SiteStatus) %>%
  summarise(mean_specialization = mean(d, na.rm = TRUE))
```

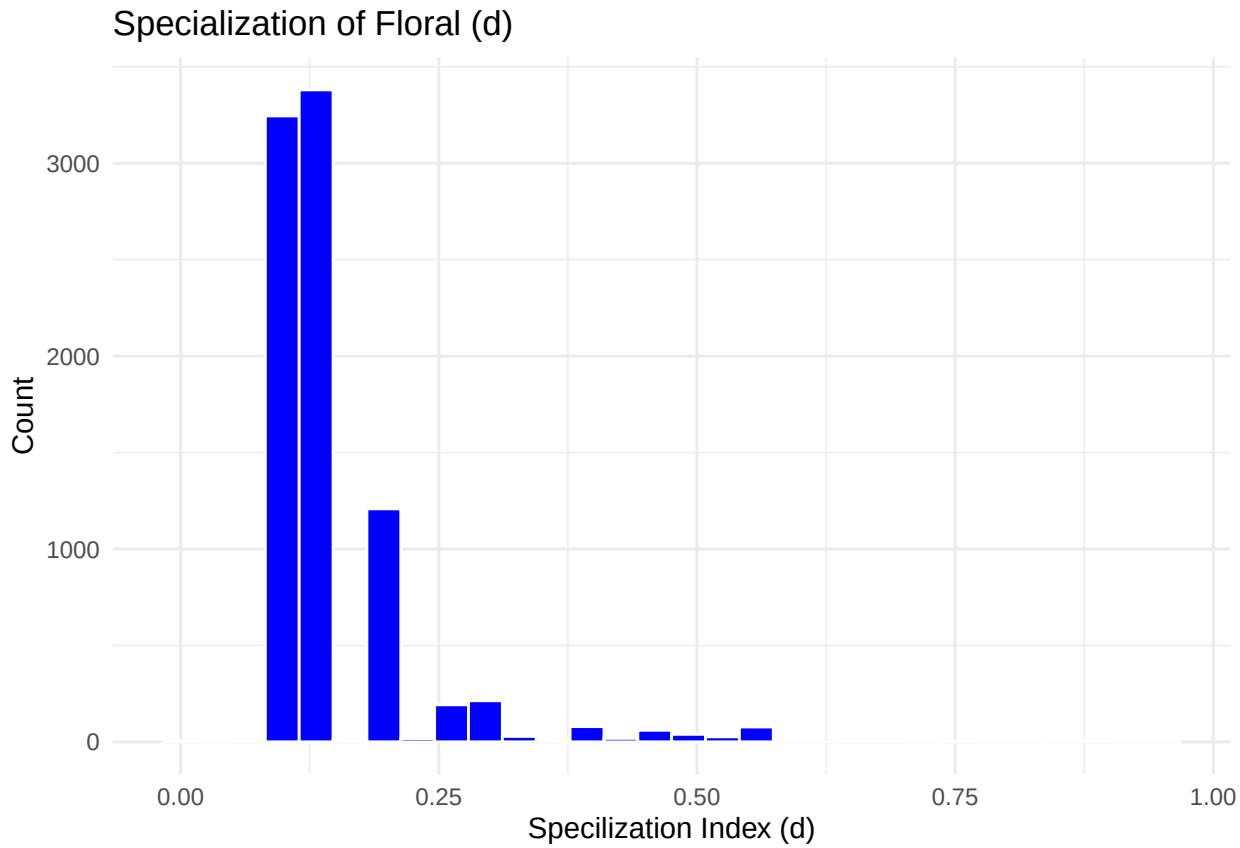
```
## # A tibble: 2 x 2
##   SiteStatus mean_specialization
##   <chr>                    <dbl>
## 1 control                  0.148
## 2 mature                   0.183
```

```
spec.traits <- spec.traits %>%
  mutate(
    Specialization = ifelse(d > 0.5, "Specialist", "Generalist")
  )

### your code here
ggplot(spec.traits, aes(x = d)) +
  geom_histogram(fill = "blue", color = "white") +
  theme_minimal() +
  labs(title = "Specialization of Floral (d)",
       x = "Specilization Index (d)",
       y = "Count")
```

```
## `stat_bin()` using `bins = 30`. Pick
```

```
## better value ‘binwidth‘.
```



- 3c. Describe what you see. – ANSWER: floral index specilization. we see that specilization if more concentrated(which it is), towards 0, most specialists are generalists. more concentrated towards one means some specialists. mainly generalist species, with a couple of specialists stragglers.

Demo: Do the bees at hedgerows and weedy field margins (controls) have the same distribution of traits?

- Null hypothesis: Mature hedgerows and weedy field margins (controls) support bees with similar traits, so the average trait values at hedgerows and controls is similar.
- Alternative hypothesis: The traits at mature hedgerows and controls are not the same.

a) Mature hedgerows have large bees and more specialized bees than would be expected if we were sampling randomly from the regional species pool.
b) Controls have smaller bees and more generalized bees than would be expected if we were sampling randomly from the regional species pool.

In the study, we tested the hypothesis above by creating a null distribution of average trait values by shuffling individuals across sites within a year. This enabled us to constrain species abundance and richness at a site while changing the identity of the individuals. Creating this type of custom null model is a bit out of the scope of what we know how to do at this point in the class, so instead we will simplify our problem by shuffling the treatment columns (mature hedgerow vs. control) across sites within a year. We will then take the average trait value for the randomized data to calculate the null expectation of controls b

Step 0: Let's first try to do our randomization with 1 year of data then work out way up.

```r
## Let's pick 2013, it was a nice year.
year1 <- sys.traits %>%
  filter(Year == 2013)

head(year1)
```

Step 1: Shuffle the treatment column

```r
## step 1) year1.sample shuffles a column

year1.shuffle <-  sample(year1$SiteStatus,
                         nrow(year1), replace=FALSE)

year1.shuffle
```

Step 2: Add the shuffled column back on to the table

```r
year1$ShuffleSiteStatus <-  year1.shuffle
head(year1)
```

Step 3: Take the mean of the simulated data.

```r
shuffle.traits <- year1 %>%
  group_by(Year, ShuffleSiteStatus) %>%
  summarise(mean_body_size = mean(n_mean_bodysize, na.rm = TRUE))
```

- 4b. Test out your function on a 2013. Run your function a few times. Do the mean values for the traits change? Why?

```r
### your code here
set.seed(123)
year1.shuffle <- sample(year1$SiteStatus, nrow(year1), replace = FALSE)

year1.shuffle
```

```
##    [1] "mature"  "control" "control"
##    [4] "control" "control" "control"
##    [7] "control" "control" "mature"
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##  [ reached 'max' / getOption("max.print") -- omitted 1470 entries ]
```

– ANSWER: because sample() randomly reassigns group labels SiteStatus changes which indiv. go not each group, changing group averages. ### Question 5 Now we want to use our function on each year of data and save our results. The perfect situation for a for loop!

- 5a. We will want to loop over the different years in the data. Start by creating a vector of the unique year values, call it 'years'. It would be nice to sort that vector as well so that the years are in order.

```
### your code here
years <- sort(unique(sys.traits$Year))
```

- 5b. We will do this together in class, but first try to think it through.

The next step was a bit tricky. Unlike before in class (lab 1), where we saved the calculations from our function into a vector, we have a dataframe I found that the easiest way to get the column names right was to start to the first year of data, then rbind() (row bind) that to the previous year. Call your randomized data random.traits

HINT: If you do as I suggested above, you will not want to include year one in your for loop. If you have a vector x, x[-1] will drop the first element of that vector.

```
random.traits <- randomizeTraits(yr = years[1], data = sys.traits)
for (yr in years[-1]) {
  temp <- randomizeTraits(yr = yr, data = sys.traits)
  random.traits <- rbind(random.traits, temp)
}
```

- 5c. Take the mean body size and specialization for each site status across years of that simulated full dataset, call it mean.years

```
mean.years <- random.traits %>%
  group_by(ShuffleSiteStatus) %>%
  summarise(
    mean_body_size = mean(mean_body_size))

mean.years
```

```
## # A tibble: 2 x 2
##   ShuffleSiteStatus mean_body_size
##   <chr>                      <dbl>
## 1 control                     1.23
## 2 mature                      1.21
```

- 5d. Now combine all the steps above into a single function called repRandomComm that:
- shuffles year 1 using the randomizeTraits function
- loops over the remaining years, row binding the data to prior year each time
- takes the mean of the traits across years, and returns that value.

It should have two arguments years and sys.traits

NOTE: It isn't A+ to have argument names be the same as objects in your Global Environment (can you think of why?), but sometimes argument names can just too silly if you keep trying to tweak them. Let's just be kinda A- with our naming to avoid confusion.

```r
### your code here
repRandomComm <- function(years, sys.traits){
  ## This function 1) shuffles year 1 using randomizeTraits function
  ## 2) loops over the remaining years, row binding the data priod year each time
  ## 3) takes the mean of the traits across years, and returns the value

  ## shuffle year 1
 random.traits <- randomizeTraits(yr = years[1], data = sys.traits)

  ## rbind on the other years
 for (yr in years[-1]) {
  new.year <- randomizeTraits(yr = yr, data = sys.traits)
  random.traits <- rbind(random.traits, new.year)
}
  ## take the mean across years
  mean.years <- random.traits %>%
    group_by(ShuffleSiteStatus) %>%
    summarise(mean_body_size=mean(mean_body_size))

  return(mean.years)
}
```

Try out your function!

```r
### your code here
repRandomComm(years, sys.traits)
```

```
## # A tibble: 2 x 2
##   ShuffleSiteStatus mean_body_size
##   <chr>                      <dbl>
## 1 control                     1.22
## 2 mature                      1.24
```

- 5d. Last step! Run your randomization 100 times using a for loop. Similar to our above for loop, it was helpful to run the randomization once to get the format of the dataframe correct, then just rbind the rest of the iterations onto that.

```r
niter <- 100

## iteration I
randomized.comms <- repRandomComm(years, sys.traits)

## loop over the rest of iterations
for(i in 1:(niter-1)){
  new.comm.means <- repRandomComm(years, sys.traits)
  randomized.comms <- rbind(randomized.comms, new.comm.means)
}

### The dimensions should be 200 3 (two rows are created each iteration)
dim(randomized.comms)
```

```
## [1] 200   2
```

Demo: Interpreting the results

To determine whether our randomized communities have similar trait values as our observed communities, we will calculate an empirical p-value.

Empirical p-value is calculated as the number of permutation tests with a 'more extreme' (larger or smaller, depending on the null hypothesis) observed test statistic.

In our case, our test statistic is the mean body size (MeanITD) and mean specialization (d).

We will start by calculating our observed test statistic for both traits.

```
obs.means.yr <- sys.traits %>%
  group_by(Year, SiteStatus) %>%
  summarise(mean_body_size = mean(n_mean_bodysize), .groups = "drop")

obs.means <- obs.means.yr %>%
  group_by(SiteStatus) %>%
  summarise(mean_body_size = mean(mean_body_size))

obs.means
```

To visualize our observed data in relation to our randomized data, we will plot a histogram of our simulated data and add a vertical line for our observed data.

```
ggplot(randomized.comms, aes(x=mean_body_size, fill=ShuffleSiteStatus)) +
  geom_histogram() +
    geom_vline(obs.means,
    mapping =aes(xintercept = mean_body_size, linetype = SiteStatus))
```

Our observed data looks quite different than our randomized data!

If our null hypothesis that bees at mature hedgerows and controls have similar traits is true what is the probability we get bees as small as we did at the controls? Let's calculate that probability. We need to calculate how many times we got values equal to or more extreme than our observed value when we randomized the treatments (mature hedgerow, control) in the data.

Remember that we can sum the TRUEs.

```
## The base R way
sum(randomized.comms$mean_body_size[
  randomized.comms$ShuffleSiteStatus == "control"] <= obs.means$mean_body_size[obs.means$SiteStatus ==
  nrow(randomized.comms)

## or the tidyverse way
subsetted.randomized <- randomized.comms %>%
  filter(ShuffleSiteStatus == "control") %>%
  select(mean_body_size)

obs.body.size <- obs.means %>%
  filter(SiteStatus == "control") %>%
  select(mean_body_size)

sum(subsetted.randomized <= obs.body.size$mean_body_size)/
  nrow(subsetted.randomized)
```

This is our empirical p-value. That is a pretty low probability that we would get our observed statistic if the null hypothesis (the traits in mature and control communities the same) was true.

We generally reject our null hypothesis if our p-value is less than 0.05. We can therefore reject our null hypothesis.

Now for the mature communities:

```
sum(randomized.comms$mean_body_size[
  randomized.comms$ShuffleSiteStatus == "mature"] >= obs.means$mean_body_size[obs.means$SiteStatus == "
  nrow(randomized.comms)
```

Again, that is a pretty low probability that we would get our observed statistic if the null hypothesis (the traits in mature and control communities the same) was true. Our p value is again less than 0.03. We can therefore reject our null hypothesis.

Question 6

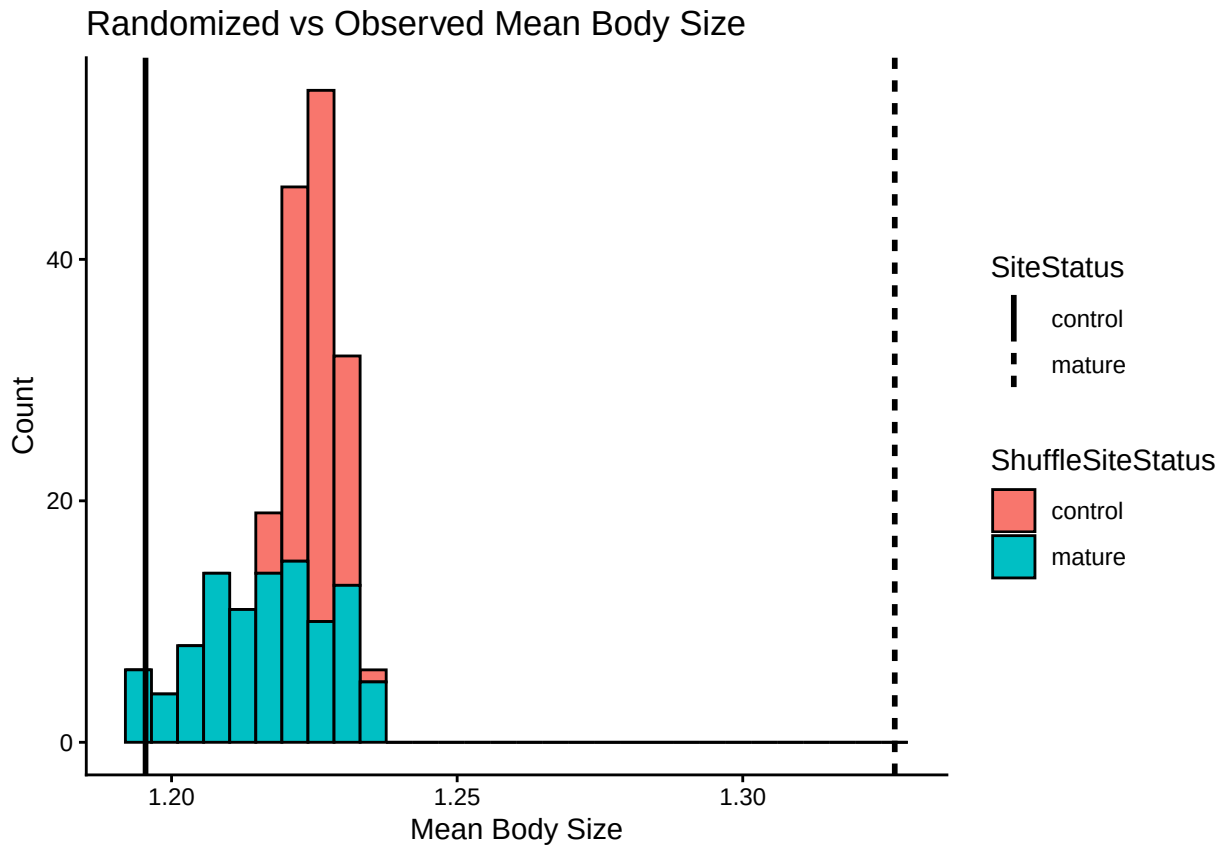
Follow the same steps for the specialization (d) - 6a. Make a histogram with the randomized and observed values.

```
library(ggplot2)

### your code here
ggplot(randomized.comms, aes(x = mean_body_size, fill = ShuffleSiteStatus)) +
  geom_histogram(color = "black") +
  geom_vline(data = obs.means,
             aes(xintercept = mean_body_size, linetype = SiteStatus),
             color = "black", size = 1) +
  labs(x = "Mean Body Size",
       y = "Count",
       title = "Randomized vs Observed Mean Body Size") +
  theme_classic()
```

```
## Warning: Using `size` aesthetic for lines was
## deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8
## hours.
## Call
## `lifecycle::last_lifecycle_warnings()`
## to see where this warning was
## generated.

## `stat_bin()` using `bins = 30`. Pick
## better value `binwidth`.
```



- 6b. What is the probability that bees at controls are as generalized (so small values of d) as what we observed, given then null hypothesis is true?

```
### your code here
prob_control <- randomized.comms %>%
  filter(ShuffleSiteStatus == "control") %>%
  summarise(prob =mean(mean_body_size <= obs.means$mean_body_size[obs.means$SiteStatus == "control"]))

print(prob_control)
```

```
## # A tibble: 1 x 1
##    prob
##   <dbl>
## 1     0
```

- 6c. Do we accept or reject our null hypothesis? – ANSWER: reject null hypothesis, observed mean body size highly unlikly to occur by random chance, signifcant effect of site status on specialization.
- 6d. What is the probability that bees at mature hedgerows are as specialized (so large values of d) as what we observed, given then null hypothesis is true?

```
prob_mature <- randomized.comms %>%
  filter(ShuffleSiteStatus == "mature") %>%
  summarize(prob = mean(mean_body_size >= obs.means$mean_body_size[obs.means$SiteStatus == "control"]))

  print(prob_mature)
```

```
## # A
## #   tibble:
## #   1
## #   x
## #   1
```

```
## Error in is.finite(width): default method not implemented for type 'list'
```

- 6f. Do we accept or reject our null hypothesis?

– ANSWER: fail to reject null hypothesis. this suggests no significant effect of site status. With these analyses we have approximated the analysis of Ponisio et al. 2016. Congratulations :)